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Comparative analysis of volatility proxies and regime-based asset allocation☆

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ABSTRACT

High uncertainty and increasing interconnectivity in global financial markets challenge static asset allocation strategies, particularly in emerging economies. This study evaluates the effectiveness of three volatility proxies (conditional, implied, and historical) in identifying market regimes through a Hidden Markov Model and examines their economic relevance for ETF allocation in Brazil and the United States using daily data from 2018 to 2024, with portfolio evaluation focused on the 2020–2024 out-of-sample period. The results indicate that conditional volatility estimated via GARCH provides more stable and operationally reliable regime identification. Model selection based on the Bayesian Information Criterion supports a three-regime specification (low, medium, and high volatility), allowing the model to capture not only extreme market states but also transitional phases often overlooked in binary frameworks. A key contribution of the study is the identification of an international risk transmission mechanism. Granger causality tests applied to regime probabilities reveal that increases in the probability of high-volatility regimes in the U.S. market precede regime shifts in the Brazilian market, suggesting that U.S. volatility regimes function as an early warning system for emerging markets. In asset allocation, the dynamic regime-based strategy outperforms static mean–variance optimization (Single Regime) in risk-adjusted terms in Brazil while successfully mitigating severe losses in the U.S. market. However, the dynamic strategy does not outperform the naive equal-weighted (1/N) benchmark, highlighting the trade-off between dynamic adaptability and estimation error under high-correlation environments. Overall, the findings suggest that regime-based strategies are best interpreted as risk-management tools rather than universal return-enhancing solutions.

1. Introduction

The intensification of global economic vulnerability and the recurrence of financial turbulence episodes have significantly increased the exposure of investment portfolios to systemic risks, reinforcing the concerns of investors and professional managers. Recent events, such as the COVID-19 pandemic, abrupt monetary tightening cycles, and geopolitical shocks, have highlighted the limitations of static allocation strategies, particularly in environments where volatility dynamics and market interconnections evolve rapidly (Bekaert & Hoerova, 2014). This challenge is particularly pronounced in emerging economies, whose higher sensitivity to external shocks intensifies risk fluctuations and persistently alters asset correlation structures.

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Recent empirical evidence indicates that economic uncertainty and changes in risk aversion play a central role in the formation of volatility regimes and the co-movement between equity markets (Asgharian et al., 2023; Bai & Cai, 2024). In this context, the ability to identify, interpret, and anticipate transitions between volatility regimes becomes an essential element of efficient risk management and capital preservation. Allocation strategies that ignore this conditional dependence tend to exhibit substantial performance deterioration precisely during periods of greatest financial stress.

Despite this recognition, a significant portion of the literature on asset allocation remains anchored in predominantly static models, which implicitly assume stability in return distributions and correlation patterns over time. These approaches neglect the intrinsic relationship between asset performance and structural changes in market conditions, limiting their risk adaptation capability (Kolm et al., 2014; Lewin & Campani, 2020; MacLean et al., 2022; Nystrup et al., 2019; Tu, 2010).

To overcome these limitations, the literature has widely turned to Hidden Markov Models (HMM) as a flexible framework for capturing regime-switching behaviors in financial markets (Ang & Timmermann, 2012; Davoodi et al., 2024; Nystrup et al., 2015). By allowing for abrupt changes in the distributional properties of returns and volatility, HMMs offer a more realistic representation of financial risk (Oprisor & Kwon, 2021). However, the effectiveness of these models critically depends on the choice of the observable variable used to infer latent states. In practice, different volatility proxies (historical, conditional, or implied) are often adopted without a systematic evaluation of how this choice affects the empirical stability of the identified regimes and the economic utility of the generated signals.

This gap becomes even more relevant in a globalized financial system, where volatility regimes can propagate asymmetrically between developed and emerging markets. Although the literature on financial contagion widely documents the international transmission of volatility shocks (Asgharian et al., 2023), relatively little attention has been devoted to analyzing how regime probabilities, inferred by structural change models, are transmitted across markets. Assessing whether the probabilistic dynamics of regimes in a dominant market, such as the United States, precede or influence regime formation in emerging markets, such as Brazil, has direct implications for systemic risk monitoring and international diversification strategies.

Given this context, this study investigates the dynamics, empirical stability, and economic relevance of volatility regimes in equity markets through a rigorous comparison of three volatility proxies (historical, conditional and implied) within the framework of Hidden Markov Models estimated using rolling windows. The analysis considers the Brazilian and U.S. markets over the period from 2018 to 2024, characterized by high global financial instability.

The article contributes to the literature on three main fronts. First, it systematically evaluates the predictive performance and empirical stability of different volatility proxies in identifying regimes, validating a three-regime specification to capture transitional market phases. The results suggest evidence favoring the use of conditional volatility, particularly in terms of robustness in classifying latent states. Second, it investigates an international risk transmission mechanism based on regime probabilities, utilizing Granger causality tests and Impulse Response Functions applied to smoothed probabilities. The results indicate that volatility dynamics in the United States act as an early warning system for the Brazilian market. Third, it assesses the economic viability of using volatility regimes in asset allocation under realistic frictions, such as transaction costs and portfolio turnover. Although the regime-based dynamic strategy presents better risk control compared to static mean–variance optimization, the results highlight a relevant trade-off between dynamic adaptation and estimation errors.

Collectively, the findings reinforce the importance of integrating regime modeling, international risk transmission, and economic evaluation under real frictions into the contemporary analysis of asset allocation strategies.

2. Literature review

The literature on portfolio management has evolved substantially since the seminal contribution of Markowitz (1952), which established the mean–variance framework as the cornerstone of Modern Portfolio Theory. By formalizing the trade-off between expected return and risk through second moments, this approach provided a tractable and intuitive basis for portfolio construction. Despite its enduring influence and widespread practical adoption, the classical mean–variance framework relies on restrictive assumptions, notably the stability of return distributions and correlations over time, and is known to be highly sensitive to estimation error, particularly in finite samples (DeMiguel et al., 2009; Michaud, 1989).

These limitations become especially pronounced in environments characterized by time-varying volatility, changing correlations, and shifts in risk premium. Empirical evidence suggests that financial markets do not operate under a single, stable data-generating process, but rather alternate between distinct structural states associated with periods of tranquility, transition, and crisis (Ardia et al., 2018; Bae et al., 2014; Oliveira et al., 2018). In such settings, static allocation strategies, including optimized mean–variance portfolios, may underperform more parsimonious rules, such as the naive equal-weighted (1/N) strategy, which remains a robust benchmark largely due to its immunity to estimation risk (DeMiguel et al., 2009, 2024; Han, 2020).

The recognition that investment opportunities evolve over time has motivated a growing body of research on dynamic and regime-dependent portfolio strategies. Rather than assuming constant moments, these approaches explicitly account for structural changes in market conditions, allowing portfolio decisions to adapt to evolving risk environments (Kolm et al., 2014; Tu, 2010). In this context, regime-switching models have emerged as a natural framework to capture nonlinear dynamics and abrupt transitions in financial time series.

2.1. Market regimes and hidden Markov models

Among regime-switching approaches, models based on Markov processes have played a central role since the pioneering work of Hamilton (1989), who introduced a probabilistic framework for modeling unobservable economic states. Building on this foundation, Hidden Markov Models (HMMs) have been widely adopted in finance to infer latent market regimes and model transitions between them (Ang & Timmermann, 2012; Li et al., 2022; Nguyen & Nguyen, 2015; Nystrup et al., 2018; Oprisor & Kwon, 2021; Zucchini et al., 2021).

A key distinction between HMMs and conventional linear time-series models lies in the assumption that observable returns are governed by an unobserved stochastic process. Rather than imposing a single distribution on the data, HMMs interpret the unconditional distribution of returns as a finite mixture of regime-dependent conditional distributions (Bartolucci et al., 2025). This structure provides substantial flexibility to accommodate salient stylized facts of financial returns, including volatility clustering, skewness, leptokurtosis, and time-varying correlations (Nystrup et al., 2016).

Furthermore, as highlighted by Fereydooni and Mahootchi (2023), the segmentation of time series into upward and downward trends, with the aid of the HMM, makes it possible to capture and incorporate the data's inherent non-linearity.

From an investment perspective, the ability of HMMs to identify and forecast regime probabilities has important implications for portfolio management. Empirical studies consistently document that conditioning portfolio decisions on inferred market regimes improves performance by enabling dynamic adjustments of asset weights to regime-specific risk–return characteristics (Davoodi et al., 2024; Kritzman et al., 2012; Nystrup et al., 2019; Oprisor & Kwon, 2021). In particular, regime-aware strategies have been shown to reduce drawdowns and enhance risk-adjusted returns relative to static allocation rules, especially during periods of heightened market stress (Bae et al., 2014; Fons et al., 2021; MacLean et al., 2022; Nystrup et al., 2018).

Nevertheless, the effectiveness of HMM-based strategies is sensitive to key modeling choices. A critical dimension concerns the selection of variables used to infer latent regimes, which can convey different information about market conditions and investor expectations. Consequently, identifying which volatility proxies most effectively capture regime dynamics remains an open empirical question, with direct consequences for the economic value of regime-based asset allocation strategies.

2.2. Volatility proxies and regime identification

Despite the widespread adoption of HMMs in the financial literature, the effectiveness of these models depends on the choice of the observable variable used to infer latent states. The literature presents a theoretical tension regarding the ideal proxy: the balance between the information content of the measures and the stability of the signal emitted by them.

Authors who advocate implied volatility argue that it encompasses superior forward information, incorporating both market expectations and the variance risk premium (Bekaert & Hoerova, 2014). However, distinct from econometric projections, implied volatility often exhibits a persistent bias over realized volatility due to this risk premium (Nystrup et al., 2016). Consequently, HMMs applied to indices such as the VIX are often praised for their ability to detect regime changes before they fully materialize in price action (Nystrup et al., 2016).

On the other hand, conditional volatility models (e.g., GARCH) are frequently cited for their robustness. While lacking the forward-looking nature of derivatives, GARCH-type models filter out transient noise, capturing volatility persistence more efficiently in regime-switching frameworks (Ardia et al., 2018). Recent evidence suggests that in structurally unstable environments, the empirical stability provided by dynamic model adaptation is a stronger determinant of out-of-sample performance than the theoretical advantage of implied measures (Bai & Cai, 2024).

This debate is particularly critical in emerging economies like Brazil. Unlike highly liquid developed markets, these environments often suffer from structural illiquidity, causing implied volatility indices to incorporate significant "liquidity noise" rather than pure risk sentiment. Consequently, relying on such proxies can trigger spurious regime transitions and excessive rebalancing, raising transaction costs that may erode the economic benefits of dynamic allocation (Lewin & Campani, 2020; Li et al., 2022). This study addresses this gap by systematically evaluating whether alternative volatility measures, characterized by lower informational noise, provide a more reliable and cost-effective guide for regime-based asset allocation (Ardia et al., 2018).

2.3. International transmission of volatility regimes

The dynamics of volatility regimes become even more complex when considering the interdependence between developed and emerging markets. The international finance literature widely documents that uncertainty and volatility shocks originating in major financial centers tend to propagate internationally, affecting returns, correlations, and risk premiums in peripheral markets (Asgharian et al., 2023; Khan, 2025).

During periods of global stress, correlations across assets and markets increase substantially, undermining the benefits of international diversification precisely when it is most needed (Bae et al., 2014). While the traditional spillover literature focuses on the transmission of returns or volatility levels, recent studies suggest that regime-based approaches provide a complementary perspective for understanding systemic risk propagation (Lewin & Campani, 2024).

Complementing this literature, the present study shifts the focus to the transmission of regime probabilities inferred by HMMs. This approach allows evaluating whether the probabilistic dynamics of latent states in a dominant market statistically precede regime formation in emerging markets, providing a more refined reading of financial contagion and the role of central markets in leading the global volatility cycle.

Table 1
Descriptive statistics of volatility proxy series.

	Series	Mean	Std. Dev.	Skewness	Kurtosis
BR	Implied Volatility (VIXBR)	0.015	0.006	3.92	30.80
	Historical Volatility (Ibovespa)	0.013	0.010	5.00	32.11
	Conditional Volatility (GARCH)	0.014	0.008	5.23	34.90
US	Implied Volatility (VIX)	0.013	0.005	2.52	14.13
	Historical Volatility (S&P500)	0.011	0.008	3.88	22.54
	Conditional Volatility (GARCH)	0.011	0.008	4.23	27.69

Note: BR stands for Brazil and US for United States. Statistics calculated on daily volatility for the out-of-sample portfolio evaluation period (2020 to 2024). Implied volatility series were annualized and converted to daily frequency.

2.4. Economic viability and transaction costs

Despite growing empirical support for regime-based strategies, recent literature emphasizes that statistical superiority does not necessarily imply economic viability. DeMiguel et al. (2024) demonstrate that while active volatility management improves gross performance metrics, such as the Sharpe Ratio, the high turnover associated with these strategies can erode net gains in the presence of transaction costs.

Complementarily, Li et al. (2022) show that the explicit incorporation of transaction costs and turnover penalties in multi-period optimization frameworks is fundamental for active management to consistently add economic value. Thus, the economic evaluation of regime-based strategies must necessarily consider realistic frictions and disciplined rebalancing rules.

In this context, the contribution of this study lies at the intersection of these strands of literature. By comparing volatility proxies within HMMs, investigating international regime transmission via state probabilities, and evaluating the economic viability of strategies under transaction costs, the article seeks to advance the understanding of when and how regime-based approaches can effectively improve asset allocation decisions in environments of high uncertainty.

3. Methodology

This section describes the data, volatility proxies, regime identification procedure, and the portfolio allocation framework. The empirical strategy is organized into three stages: (i) construction of volatility proxies and sample definition; (ii) identification and forecasting of volatility regimes using Hidden Markov Model (HMM) estimated in rolling window; and (iii) construction and evaluation of dynamic portfolios conditioned to the predicted regimes, adjusted for transaction costs.

3.1. Data and proxies for volatility

We used daily returns from the Ibovespa (Brazil) and S&P 500 (United States) indexes covering the period from January 2018 to December 2024. This interval encompasses episodes of high instability, including the COVID-19 pandemic, a global monetary tightening cycle, and geopolitical crises, providing a suitable environment for analyzing volatility regimes.

The initial interval, from January 2018 to December 2019, was used exclusively as an in-sample training period for model structure calibration, specifically for selecting the number of latent regimes. The rolling window estimation begins in January 2019, while the portfolio evaluation stage (backtest) considers the period from June 2020 to December 2024, preserving the preceding interval as a statistical burn-in period. This procedure ensures both the stable convergence of the estimation algorithm and the proper formation of the largest rolling windows considered.

Three alternative volatility proxies are considered. First, historical volatility is computed as the rolling standard deviation of daily returns over a 21-trading-day window, approximating one trading month. Second, conditional volatility is estimated using a GARCH(1,1) specification, capturing volatility persistence and clustering commonly observed in financial returns and implemented using the `rugarch` package in R (Ghalanos, 2023). Third, implied volatility is proxied by market-based volatility indices, reflecting forward-looking expectations embedded in option prices. For the United States, the VIX index (CBOE) was used. For Brazil, a composite series was constructed using IVol-BR (NEFIN, 2017) and, for the most recent period, VXBR (S&P Dow Jones Indices, 2024), ensuring the necessary temporal coverage.

Table 1 presents the descriptive statistics of the volatility proxies for the Brazilian and U.S. markets. All measures exhibit pronounced positive skewness and high kurtosis, indicating leptokurtic distributions with heavy tails.

The presence of these stylized facts motivates the modeling strategy adopted. Although asset returns may follow approximately Gaussian distributions conditioned on a given latent state, the unconditional distribution, observed without regime conditioning, emerges as a finite mixture of regime-specific Gaussian components. This mixture structure naturally generates heavier tails and excessive kurtosis relative to a single normal distribution, thus capturing the non-linear dynamics and extreme events commonly observed in financial returns. Consequently, the pronounced deviations from normality observed in the data highlight the limitations of traditional linear models and justify the adoption of Hidden Markov Models (HMMs), which explicitly accommodate regime-dependent dynamics in both mean and volatility processes (De Franco et al., 2017; Nystrup et al., 2019; Peng et al., 2022).

In the allocation stage, Exchange-Traded Funds (ETFs) representing different factors and sectors of each market were selected (see Table 2), with assets denominated in their local currencies. The choice of ETFs stems from their characteristics, which combine

Table 2
Description of finance metrics being analyzed.

Variable	Market	Description
Exchange-Traded Funds (ETFs):		
BOVA11	BR	Tracks the performance of the Ibovespa index.
SMAL11	BR	Tracks the performance of the Brazilian Small Caps index.
IMAB11	BR	Replicates Brazilian inflation-linked government bonds.
FIXA11	BR	Composed of Brazilian fixed-rate government bonds.
DIVO11	BR	Tracks the IDIV, a Brazilian dividend index.
FIND11	BR	Tracks the IFNC, an index of the Brazilian financial sector.
MATB11	BR	Tracks the IMAT, a Brazilian basic materials index.
GOVE11	BR	Tracks the IGCT, a Brazilian corporate governance index.
IVV	US	Tracks the performance of the S&P500 index.
IWM	US	Replicates U.S. Small Caps stocks - Russell 2000 index.
IEF	US	Replicates medium-term fixed-rate U.S. Treasury bonds.
TIP	US	Composed of U.S. Treasury Inflation-Protected Securities.
DVY	US	Tracks high-dividend U.S. companies.
FTXO	US	Tracks U.S. companies in the financial sector.
VAW	US	Tracks U.S. companies in the basic materials sector.
ESGU	US	Tracks U.S. mid/large-cap ESG stocks.
Risk-Free Rate:		
Selic	BR	Short-term interest rate set by the Central Bank of Brazil.
IRX	US	Interest rate of the 13-week U.S. Treasury Bills (T-bills).

Note: BR refers to data from the Brazilian market (in BRL); US refers to data from the United States market (in USD). All data are used daily. Returns are computed as log-returns from closing prices. The S&P 500, Ibovespa, ETFs, VIX, and IRX series were obtained from Yahoo Finance, and the Selic rate was retrieved from the Central Bank of Brazil.

diversification, liquidity, and accessibility. The number of ETFs considered is constrained by the availability of long, liquid series in the Brazilian market, which has expanded primarily in recent years.

The Brazil–United States empirical design is intentional, contrasting an emerging market, characterized by greater financial frictions and increased sensitivity to external shocks, with a developed market, which is more solid and liquid (Asgharian et al., 2023). Although broad transnational panels are recurrent in the literature on financial contagion, the focus on this specific bilateral relationship allows for a more precise isolation of the mechanisms of volatility transmission, particularly the Early Warning System hypothesis, according to which uncertainty shocks in core markets anticipate regime changes in peripheral markets.

Furthermore, the symmetry in asset selection (8 ETFs per market to cover factors and sectors) provides a controlled environment to assess the stability of regimes. Crucially, the restriction to this specific asset universe is driven by strict data availability: these are the primary liquid Brazilian ETFs with continuous daily trading over the full 5-year sample.

3.2. Identification of volatility regimes via hidden Markov models

Volatility regimes are identified using a Hidden Markov Model (HMM), in which the observable variable Y_t, representing a volatility proxy, is governed by an unobserved first-order stochastic process $S_t \in \{1, \dots, k\}$. The latent state process follows a homogeneous Markov chain characterized by a transition probability matrix $\boldsymbol{A}$. Conditional on regime $S_t = j$, the volatility proxy is assumed to follow a Gaussian distribution, $Y_t | S_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2)$. Model estimation is implemented in R using the `depmix` package (Visser & Speekenbrink, 2010).

The optimal number of states k is determined empirically using the Bayesian Information Criterion (BIC). The criterion is evaluated over the training period (2018–2019) for model specifications with $k \in \{2, 3, 4, 5\}$. The results indicate that a three-regime structure provides the best trade-off between statistical fit and parsimony. Economically, these regimes are interpreted as low, medium, and high volatility states.

The inclusion of an intermediate regime allows the model to capture transitional phases characterized by elevated uncertainty without full-scale crisis dynamics, which are frequently observed during monetary tightening cycles and post-crisis normalization periods.

To mitigate look-ahead bias and accommodate potential structural instability in volatility dynamics, the Hidden Markov Model (HMM) parameters $\theta = \{\mu_j, \sigma_j^2\}_{j=1}^k$ and the transition matrix $\boldsymbol{A}$ are estimated recursively using a rolling window approach.

As illustrated in Fig. 1, the model is trained on fixed-length windows $W \in \{126, 252, 504\}$ trading days and produces a one-step-ahead forecast at time $t + 1$ before the window advances by one observation.

At each iteration, the expected volatility forecast is defined as the conditional expectation given the information set $\mathcal{F}_t$,

$$\hat{y}_{t+1} = \mathbb{E}[Y_{t+1} \mid \mathcal{F}_t] = \sum_{j=1}^{k} P(S_{t+1} = j \mid S_t)\, \hat{\mu}_{j,t}, \tag{1}$$

where $\hat{\mu}_{j,t}$ denotes the regime-specific mean estimated over the rolling window ending at time t, and $P(S_{t+1} = j \mid S_t)$ represents the predictive regime probabilities derived from the filtered probabilities and the estimated transition matrix.

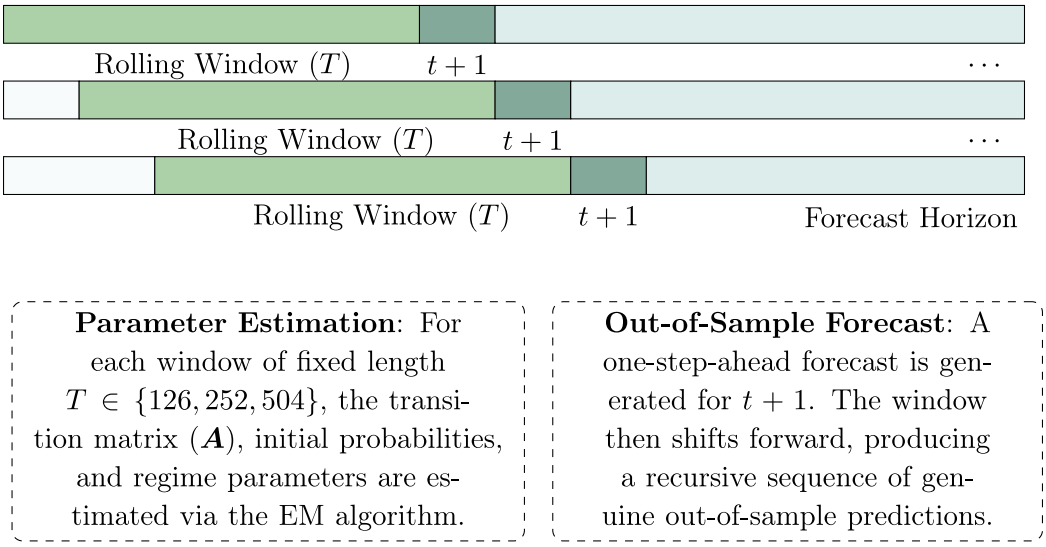


Fig. 1. Rolling window estimation and one-step-ahead forecasting scheme for the HMM.

The selection of the volatility proxy used in the portfolio allocation stage is based on a dual criterion: *predictive accuracy* and *parameter stability*.

Forecast accuracy is evaluated strictly out of sample using the Mean Absolute Error (MAE) and the Root Mean Squared Error (RMSE).

However, similar forecast errors across proxies do not necessarily imply comparable economic reliability, as noisy volatility signals may generate unstable regime classifications and excessive portfolio turnover.

Therefore, proxy selection additionally prioritizes temporal stability of the estimated HMM parameters. Specifically, we assess the dispersion of regime-specific moments (μ_j, σ_j) and regime persistence probabilities (a_{ii}) across rolling windows using the coefficient of variation (CV).

Lower CV values indicate more stable regime dynamics and more reliable probabilistic signals for portfolio decisions.

To ensure a consistent economic interpretation and address the label-switching problem inherent to regime-switching models, regimes are reordered in each rolling window according to ascending volatility levels, $\sigma_1 < \sigma_2 < \sigma_3$, corresponding to low, medium, and high-volatility states.

Although the HMM yields probabilistic forecasts of future regimes, the operational implementation of regime-based strategies requires a discrete decision rule. In this study, portfolio rebalancing is triggered using a threshold rule applied to the one-step-ahead predicted regime probabilities.

Let $\pi_{t+1}(j) = P(S_{t+1} = j \mid \mathcal{F}_t)$ denote the predicted probability of regime j at time $t+1$.

The operational implementation of the regime-based strategy relies on identifying the most likely future state. Formally, the predicted regime for time $t+1$ is defined by the highest predicted probability, such that $\hat{S}t + 1 = \arg\max j\pi_{j,t+1}$. Portfolio rebalancing is triggered whenever this identified dominant regime changes from the previous period. Complementarily, a probability threshold τ is introduced not as a mandatory constraint for allocation, but as a sensitivity parameter utilized specifically for the Switching Range (SR) diagnostic. This approach allows for evaluating the robustness of the signals across an uncertainty interval $\mathcal{T} = [\tau_L, \tau_U]$ without imposing a rigid probability hurdle for the core investment strategy".

Because the choice of τ is inherently arbitrary and subject to estimation uncertainty, small perturbations may lead to materially different regime classifications and rebalancing paths. To formally evaluate the operational robustness of the generated signals, we introduce two complementary diagnostics.

The Switching Range (SR) measures the sensitivity of the number of regime switches to variations in the decision threshold.

Considering an uncertainty interval $\mathcal{T} = [\tau_L, \tau_U]$ (e.g., $\tau \in [0.40, 0.60]$), let $N(\tau)$ denote the total number of regime changes observed over the evaluation sample. The SR is defined as

$$SR = \max_{\tau \in \mathcal{T}} N(\tau) - \min_{\tau \in \mathcal{T}} N(\tau). \tag{2}$$

A high SR indicates that the rebalancing frequency of the strategy is highly sensitive to small perturbations in the decision rule, reflecting low operational robustness.

The Uncertainty Range (UR) is defined as the proportion of observations in the total sample where the model lacks classification conviction. Formally, it is calculated as the sample average of an indicator function that identifies points where the difference between the two highest predicted probabilities is smaller than a margin m $(\pi^{(1)}_{t+1} - \pi^{(2)}_{t+1} \leq m)$. A margin of $m = 0.05$ was adopted to

identify periods of classification ambiguity, corresponding to cases in which competing regime probabilities differ by less than five percentage points, reflecting a conservative margin of probabilistic ambiguity.

Low values of SR and UR, combined with stable coefficients of variation for regime-specific parameters, identify the volatility proxy that delivers the most distinct, robust, and economically actionable signals for dynamic asset allocation under market frictions.

3.3. International transmission of volatility regimes

Extending the analysis beyond domestic regime identification, we investigate the international transmission of latent uncertainty between a core market (the United States) and a peripheral market (Brazil). While Asgharian et al. (2023) demonstrate that US-based uncertainty measures are strong predictors of international market volatility, we specifically test for directional causality following the framework of Khan (2025), who applies Granger causality tests to examine interlinkages among BRICS nations.

Let $P_t \equiv P(S_t = H|\mathcal{F}_t)$ denote the smoothed probability that the market is in the high-volatility regime (H) at time t, based on the full information set $\mathcal{F}_t$. These probabilities summarize the perception implicit in the model of latent market uncertainty, rather than actually observed volatility shocks.

Since P_t is strictly bounded within the unit interval $(0, 1)$, its distributional properties violate the normality assumptions required for standard Vector Autoregressive (VAR) inference. To address this, we apply a logit transformation to map the probabilities onto the real line $(-\infty, +\infty)$:

$$\tilde{P}_t = \ln\left(\frac{P_t}{1 - P_t}\right) \tag{3}$$

This transformation improves the approximation to normality, ensuring robust statistical inference in the VAR framework. The resulting specification tests whether shifts in the probability of a high-volatility regime in the developed market statistically precede regime shifts in the emerging market, providing direct evidence of volatility contagion mechanisms.

3.4. Regime-based asset allocation

The asset allocation strategy exploits the regime forecasts generated by the HMM to optimize the risk–return trade-off. Unlike approaches that rely exclusively on the most likely regime, thus neglecting regime uncertainty, we adopt a mixed distribution approach, in which portfolio moments are calculated probabilistically between regimes.

Let $\pi_{j,t+1} = P(S_{t+1} = j|\mathcal{F}_t)$ denote the predicted probability that regime j materializes at time $t+1$. The conditional moments used in portfolio optimization, the expected return vector $\hat{\boldsymbol{\mu}}_{t+1}$ and the covariance matrix $\hat{\boldsymbol{\Sigma}}_{t+1}$, are computed as probability-weighted averages across regimes:

$$\hat{\boldsymbol{\mu}}_{t+1} = \sum_{j=1}^{k} \pi_{j,t+1} \boldsymbol{\mu}_j \tag{4}$$

$$\hat{\boldsymbol{\Sigma}}_{t+1} = \sum_{j=1}^{k} \pi_{j,t+1} \left(\boldsymbol{\Sigma}_j + (\boldsymbol{\mu}_j - \hat{\boldsymbol{\mu}}_{t+1})(\boldsymbol{\mu}_j - \hat{\boldsymbol{\mu}}_{t+1})'\right) \tag{5}$$

This formulation captures not only the within-regime risk through $\boldsymbol{\Sigma}_j$, but also the jump risk arising from uncertainty about the future regime, represented by the second term in the covariance expression.

Portfolio weights are obtained by solving a mean–variance optimization problem with an explicit penalty on portfolio turnover to account for transaction costs, as recommended by Li et al. (2022):

$$\begin{aligned} \max_{\mathbf{w}_t} \quad & \mathbf{w}_t' \hat{\boldsymbol{\mu}}_{t+1} - \frac{\lambda}{2} \mathbf{w}_t' \hat{\boldsymbol{\Sigma}}_{t+1} \mathbf{w}_t - \gamma \sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}| \\ \text{s.t.} \quad & \mathbf{w}_t \geq 0, \\ & \mathbf{1}'\mathbf{w}_t = 1. \end{aligned} \tag{6}$$

The expected return vectors $\hat{\boldsymbol{\mu}}_{t+1}$ and the covariance matrix $\hat{\boldsymbol{\Sigma}}_{t+1}$ are estimated using a mixture approach, weighted by the regime probabilities at time $t + 1$. The risk aversion parameter is set at $\lambda = 1$ to isolate the effects of regime identification and transaction costs on strategy performance, avoiding interference from specific investor risk preferences.

To evaluate the robustness of the strategy against market frictions, the transaction cost levels γ are not optimized ex-post. Instead, they are analyzed through a sensitivity grid to preserve the ex-ante nature of the evaluation, comprising values of $\{0, 2.5, 5.0, 10.0, 20.0\}$ basis points (bps). These levels allow for direct economic interpretation and cover a conservative range of plausible frictions for ETFs across various market conditions.

In the optimization problem, γ penalizes turnover proportionally to the ℓ_1 norm of the weight changes, capturing linear trading costs in a reduced-form manner.

3.5. Benchmarks and performance evaluation

The performance of the proposed regime-based dynamic strategy is evaluated against two robust benchmarks:

Table 3
Comparison of volatility proxies for regime identification via HMM.

Country	Metric	Implied volatility			Conditional volatility			Historical volatility		
		126	252	504	126	252	504	126	252	504
BR	RMSE	0.0041	0.0042	0.0047	0.0037	0.0046	0.0051	0.0047	0.0058	0.0061
	MAE	0.0017	0.0018	0.0021	0.0014	0.0018	0.0018	0.0018	0.0024	0.0024
US	RMSE	0.0025	0.0030	0.0036	0.0038	0.0048	0.0053	0.0037	0.0045	0.0048
	MAE	0.0014	0.0018	0.0021	0.0020	0.0024	0.0024	0.0018	0.0022	0.0023

Note: RMSE and MAE are computed from one-step-ahead forecasts of each volatility proxy Y_{t+1} relative to its realized value, under rolling-window estimation.

1. Single-Regime (Static) Strategy: A conventional mean–variance portfolio optimized using rolling windows, but ignoring regime heterogeneity.
2. Naive 1/N Portfolio: An equally weighted portfolio with daily rebalancing, widely recognized as a challenging benchmark in out-of-sample settings (DeMiguel et al., 2009).

Performance is assessed using standard risk-adjusted metrics, including the Sharpe ratio, Sortino ratio, and Maximum Drawdown. Crucially, we also compute the Certainty Equivalent Return (CER), which allows for a utility-based comparison and formal statistical testing of performance differences between the proposed strategy and the benchmarks.

4. Empirical results

This section presents the empirical findings in three parts. First, we compare volatility proxies and HMM specifications, selecting the proxy-window combination and the number of regimes based on out-of-sample criteria and statistical parsimony. Second, we characterize the regimes (means, dispersion, and persistence) and evaluate their stability and classification robustness over time. Third, we quantify international transmission using Granger causality tests and Impulse Response Functions (IRF) applied to regime probabilities, followed by an economic performance evaluation of regime-conditioned strategies under transaction costs.

4.1. Volatility and hidden Markov model specification

The model structure was determined starting with the selection of the number of latent regimes (K). The prevailing literature on dynamic allocation often adopts two-regime models (low and high volatility), arguing that this configuration offers an appropriate balance between economic interpretability and statistical parsimony (Li et al., 2022; Ma et al., 2019; Nystrup et al., 2019). However, restricting the model to only two states may limit its ability to capture intermediate transition phases or periods of moderate uncertainty, especially in contexts characterized by severe exogenous shocks, such as the COVID-19 pandemic and recent geopolitical conflicts.

To assess this trade-off without incurring overfitting, HMM specifications with $K \in \{2, 3, 4, 5\}$ regimes were estimated. The Bayesian Information Criterion (BIC), computed over the training sample, indicates that introducing a third regime leads to a substantial reduction in the BIC for both analyzed markets, signaling a meaningful improvement in model fit after accounting for additional complexity. In contrast, specifications with four or five regimes yield only marginal BIC improvements accompanied by increased instability in parameter estimates, suggesting a heightened risk of overfitting.

Thus, the three-regime specification represents a natural extension of the traditional binary framework, allowing the identification of an intermediate volatility state associated with transition periods, while preserving statistical parsimony and model stability. This choice is consistent with recent evidence emphasizing the economic relevance of intermediate volatility regimes in high-uncertainty environments (Ma et al., 2019).

To evaluate the effectiveness of alternative volatility proxies (implied, conditional, and historical) in identifying and forecasting volatility regimes, we analyze their out-of-sample predictive performance in both markets. Forecast accuracy is assessed using the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), interpreted as predictive loss functions. These metrics measure the average deviation between one-step-ahead forecasts of each proxy, estimated using rolling windows of 126, 252, and 504 trading days, and their corresponding realized values.

Table 3 reports the resulting RMSE and MAE statistics. Although no statistically significant differences across proxies are detected according to the Kruskal–Wallis test, these results should be interpreted with caution. RMSE and MAE are not hypothesis-testing statistics, but rather summary measures of predictive accuracy. Consequently, the absence of statistically significant differences in median forecast errors does not preclude economically meaningful distinctions across proxies in terms of parameter stability, regime persistence, and the interpretability of the latent states.

Specifically, the conditional volatility proxy exhibits greater temporal stability in regime-specific moments and transition probabilities across rolling windows, a crucial feature for the operational reliability of regime-conditioned allocation strategies. While implied volatility incorporates forward-looking information, its regime classifications appear more sensitive to short-term fluctuations and potential microstructure effects, particularly in the Brazilian market.

Table 4
Stability of regime classification across probability threshold variations.

Market	Metric	Implied Volatility			Conditional Volatility			Historical Volatility		
		126	252	504	126	252	504	126	252	504
BR	SR	45	32	17	16	16	15	8	6	4
	UR	0.056	0.039	0.027	0.022	0.022	0.022	0.013	0.015	0.015
US	SR	24	12	21	18	7	13	2	2	0
	UR	0.029	0.028	0.021	0.032	0.020	0.030	0.003	0.005	0.008

Notes: Switching Range (SR) is defined as the difference between the maximum and minimum number of regime shifts obtained by varying the classification threshold within the $[0.40, 0.60]$ interval. Uncertainty Range (UR) measures the fraction of observations with ambiguous regime classification, defined by a small difference between the two highest predicted regime probabilities. Lower values indicate greater robustness.

Table 5
Coefficient of variation for HMM stay probabilities by volatility proxy and estimation window.

Market	Metric	Implied volatility			Conditional volatility			Historical volatility		
		126	252	504	126	252	504	126	252	504
BR	$CV(a_{11})$	0.186	0.083	0.042	0.051	0.033	0.021	0.044	0.027	0.025
	$CV(a_{22})$	0.191	0.081	0.041	0.053	0.032	0.021	0.042	0.025	0.024
	$CV(a_{33})$	0.199	0.089	0.041	0.058	0.034	0.022	0.055	0.030	0.021
US	$CV(a_{11})$	0.079	0.043	0.020	0.055	0.039	0.015	0.034	0.022	0.016
	$CV(a_{22})$	0.074	0.046	0.019	0.056	0.040	0.013	0.035	0.025	0.009
	$CV(a_{33})$	0.076	0.044	0.019	0.063	0.045	0.016	0.045	0.029	0.024

Note: The coefficient of variation (CV) is defined as the standard deviation divided by the mean, computed across rolling windows for each estimated stay probability a_{ii}. Lower CV values indicate greater temporal stability of the HMM transition dynamics.

Given the similarity of out-of-sample forecast errors across proxies, the final selection is therefore driven primarily by regime stability and classification robustness. These properties are assessed using the Switching Range (SR), the Uncertainty Range (UR), and the coefficient of variation of the stay probabilities ($CV(a_{ii})$).

Table 4 shows that implied-volatility-based proxies are substantially more sensitive to the choice of the classification threshold, especially in Brazil. The VIXBR proxy exhibits a pronounced Switching Range in the 126-day window (45 regime shifts) and a high Uncertainty Range (0.056), indicating frequent regime reclassifications in response to minor threshold variations.

In contrast, the conditional volatility proxy displays markedly greater robustness. In Brazil, the Switching Range remains limited (between 15 and 16 shifts) with an Uncertainty Range close to 0.022 across all estimation windows. For the U.S. market, the highest stability is observed in the 252-day window, with only seven regime switches and an Uncertainty Range of 0.020.

This evidence is reinforced by Table 5, which reports the coefficient of variation of the stay probabilities. The GARCH-based specification consistently yields lower $CV(a_{ii})$ values, particularly in the 252-day window, indicating stable transition dynamics across rolling estimations. Although historical volatility exhibits very low Switching Ranges, this behavior reflects stronger temporal smoothing, which may limit its ability to detect abrupt regime changes.

Based on the selected specification, three regimes, conditional volatility, and a 252-day estimation window, low and medium volatility regimes dominate in both markets, while the high-volatility regime is less frequent but more persistent. Fig. 2 illustrates the daily returns of the U.S. and Brazilian equity markets overlaid with the corresponding volatility regimes identified by the model.

As shown in Fig. 2, periods of pronounced market turbulence are consistently classified as high-volatility regimes. In particular, the COVID-19 shock and subsequent episodes of financial stress are clearly associated with the high-volatility state in both markets. More stable periods are captured by the low-volatility regime, while intermediate phases correspond to the medium-volatility state.

This visual evidence supports the economic interpretability of the model and confirms its ability to identify meaningful shifts in market conditions. The presence of a distinct intermediate regime further suggests that market transitions are gradual rather than abrupt, which is consistent with the dynamics of financial volatility.

Over the 2020–2024 period, the U.S. market spent more time in the high-volatility regime (303 trading days) than the Brazilian market (236 days), whereas Brazil exhibits higher regime turnover.

This pattern is consistent with the stylized fact of volatility clustering and suggests that, although emerging markets tend to experience more frequent regime switching, stress episodes may be more persistent in core markets due to their broader exposure to global shocks. Contrary to initial expectations, the U.S. market faced a more prolonged high-volatility phase during the sample period. Recent studies emphasize that economic and geopolitical uncertainty plays a central role in shaping the persistence and intensity of volatility regimes, potentially outweighing traditional emerging-versus-developed market distinctions (Asgharian et al., 2023).

The sample period encompasses not only the post-pandemic recovery but also major geopolitical events, including the war in Ukraine and rising tensions in the Middle East. Given its central position in the global financial and geopolitical system, the U.S. market may have been more directly and persistently affected by these shocks (Douglas et al., 2025).

To investigate cross-market dependence, Granger causality tests are applied to the logit-transformed probabilities of the high-volatility regime. The results indicate that the U.S. volatility state Granger-causes the Brazilian state ($F = 7.20$, $p < 0.001$ at lag 2).

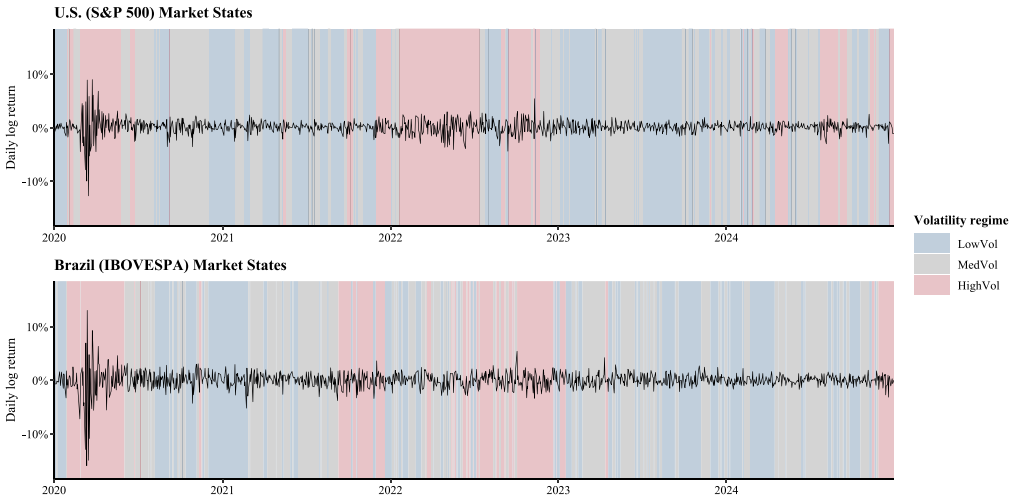


Fig. 2. Daily returns and volatility regimes estimated by the selected three-state Hidden Markov Model. The shaded areas indicate the predicted regime classification (low-, medium-, and high-volatility states).

Although reverse causality is also statistically significant ($F = 3.05$, $p = 0.047$), the relative magnitude of the test statistics reveals a clear asymmetry, identifying the U.S. as the dominant source of volatility transmission.

Impulse Response Function (IRF) analysis further clarifies the dynamics of this transmission. Exogenous shocks to the probability of a high-volatility regime in the U.S. increase the corresponding probability in Brazil with a lag of approximately two trading days. The persistence of this response suggests that the Brazilian market not only absorbs the initial shock but internalizes external risk perceptions into its own regime dynamics.

These findings support the "Early Warning System" hypothesis and align with the broader literature on international volatility spillovers. Asgharian et al. (2023) show that global risk aversion, largely originating in the U.S., is a key driver of international equity co-movements. In the Brazil–U.S. context, our results provide granular confirmation of Lewin and Campani (2020, 2024), who document that while Brazilian bull markets tend to be idiosyncratic, crash regimes are strongly synchronized with U.S. uncertainty shocks.

Taken together, these results indicate that the selected HMM specification delivers economically meaningful and statistically stable volatility regimes, while also capturing relevant international spillovers. The dominance of U.S. driven high-volatility signals and their lead–lag transmission to the Brazilian market reinforces the interpretation of regime probabilities as forward-looking indicators of systemic risk rather than mere statistical classifications.

This characterization motivates the use of regime probabilities as predictive state variables in the portfolio decision process. Conditioning asset allocation on the prevailing volatility regime allows the strategy to explicitly account for regime-dependent changes in expected returns, risk, and cross-asset dependence. Importantly, portfolio rebalancing is triggered only by changes in the inferred regime, rather than at fixed calendar intervals. This operational constraint allows the strategy to adjust to changing market conditions while curbing unnecessary turnover and transaction costs, thereby mitigating the trade-off between dynamic adaptability and estimation error.

Fig. 3 complements the formal transmission analysis by depicting the smoothed posterior probability of the high-volatility regime in the U.S. and Brazil over time. Rather than serving as independent causal evidence, the figure offers visual support for the Granger-causality and impulse-response results, which more rigorously establish the directional lead–lag relationship.

The following subsection formalizes this regime-conditioned allocation framework and evaluates its economic performance under realistic market frictions.

4.2. Using HMM for portfolio investment

In the portfolio construction stage, the investment universe was intentionally expanded to capture the heterogeneous response of assets to the identified volatility regimes. For each market, eight ETFs were selected, covering not only the broad market index and fixed-income assets but also style factors (small caps, dividends, ESG, and corporate governance) and sector-specific exposures sensitive to the economic cycle, such as financials and basic materials.

In the Brazilian market, the portfolio includes the following ETFs: BOVA11 (Ibovespa), SMAL11 (Small Caps), IMAB11 (inflation-linked government bonds), FIXA11 (pre-fixed government bonds), DIVO11 (Dividends), FIND11 (Financial sector), MATB11 (Basic Materials), and GOVE11 (Corporate Governance). This composition allows for the simultaneous capture of the market risk premium, fixed-income dynamics, and the sensitivity of typically pro-cyclical factors and sectors to regime transitions.

For the U.S. market, a methodologically symmetric structure was adopted, consisting of the following ETFs: IVV (S&P 500), IWM (Small Caps), IEF (Intermediate term Treasuries), TIP (Inflation protected Treasuries), DVY (Dividends), FTXO (Financial sector),

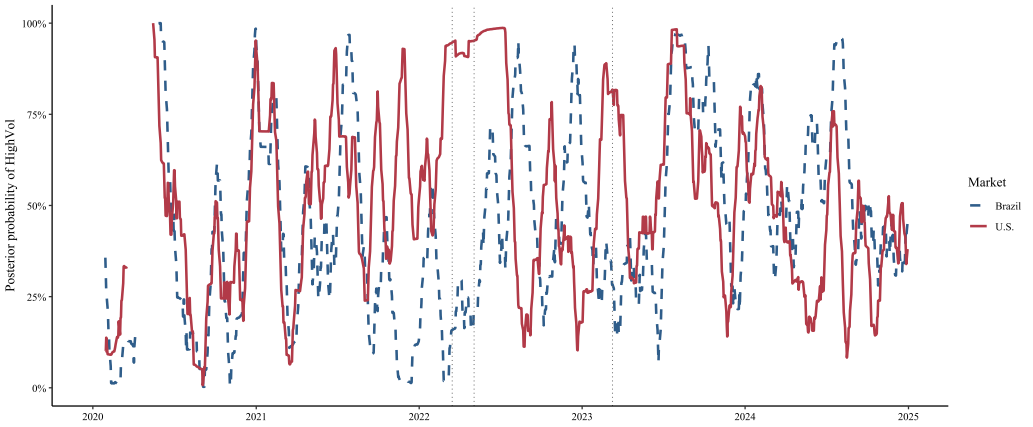


Fig. 3. High-volatility regime probabilities in the U.S. and Brazilian markets. The figure displays the smoothed posterior probability of the high-volatility regime, as inferred from the selected three-state HMM for both the U.S. and Brazil. The series are presented as 21-day rolling averages to emphasize persistent shifts in latent volatility conditions.

Table 6
ETF performance indicators by market and year (Annual Excess Returns %).

Year	Brazil							
	BOVA	SMAL	IMAB	FIXA	DIVO	FIND	MATB	GOVE
2020	0.139	−4.103	2.954	5.396	4.268	−1.064	45.36	1.537
2021	−13.94	−19.40	−6.50	−8.32	−6.89	−26.65	6.66	−14.83
2022	11.79	−9.93	4.17	6.50	16.76	14.10	−2.27	7.42
2023	28.99	22.33	15.11	15.61	29.03	38.79	10.71	26.49
2024	−6.366	−17.59	−3.227	−2.22	−0.278	−11.05	−11.98	−5.416
Full period performance (annualized)								
Excess return (%)	3.293	−7.395	2.387	3.286	8.460	0.441	8.747	2.275
Volatility (%)	26.00	32.00	9.97	7.31	21.90	28.90	29.70	26.20
Sharpe Ratio	0.127	−0.023	0.239	0.450	0.386	0.015	0.295	0.086
	United States							
Year	IVV	IWM	IEF	TIP	DVY	FTXO	VAW	ESGU
2020	11.38	10.12	9.72	10.41	−12.19	−26.41	10.36	15.27
2021	27.66	11.81	−3.46	5.56	30.16	35.54	25.36	25.75
2022	−20.52	−23.55	−15.59	−12.64	−0.00	−21.50	−14.70	−22.70
2023	25.23	14.54	3.19	3.55	0.00	−4.44	11.99	24.66
2024	24.42	8.88	−0.06	1.67	15.00	25.40	−1.05	23.78
Full period performance (annualized)								
Excess return (%)	12.00	3.23	−1.77	1.37	5.61	−1.31	5.49	11.60
Volatility (%)	21.40	27.50	7.96	7.16	22.60	36.20	24.80	21.70
Sharpe Ratio	0.559	0.117	−0.001	0.191	0.248	−0.004	0.222	0.533

Note: Full period from Jan/2020 to Dec/2024. Excess returns are calculated by subtracting the average annual risk-free rate (Selic for BR, IRX for US) from the ETF's average annual return. The Sharpe ratio follows the refinement by Israelsen (2005) for negative excess returns.

VAW (Basic Materials), and ESGU (ESG factor). The inclusion of sector-specific and thematic ETFs enhances the model's ability to exploit dynamic correlations, as sectors such as financials and basic materials tend to react differently to inflationary shocks and changes in monetary policy compared to broad market indices.

Table 6 highlights clear differences in the risk–return profiles across asset classes and markets. Over the full period (2020–2024), fixed-income ETFs exhibited substantially lower annualized volatility compared to equity ETFs, both in Brazil (IMAB11: 9.97%; FIXA11: 7.31% vs. BOVA11: 26.0% and SMAL11: 32.0%) and the United States (TIP: 7.16%; IEF: 7.96% vs. IVV: 21.4% and IWM: 27.5%).

ETFs associated with specific factors and sectors show greater return dispersion, particularly small caps and the financial sector. In Brazil, SMAL11 and FIND11 present volatilities of 32.0% and 28.9%, respectively, while in the U.S. market, the financial ETF (FTXO) reaches a volatility of 36.2%, exceeding even the broad market index.

Regarding risk-adjusted performance, aggregate Sharpe ratios in the Brazilian market remained modest on average, with superior relative performance concentrated in fixed-income and dividend ETFs. In the U.S. market, higher risk-adjusted returns were observed for broad market indices and ESG factors.

Table 7
Asset correlations across volatility regimes and full sample.

Brazilian Market					U.S. Market				
Asset Pairs	High	Med.	Low	Full	Asset Pairs	High	Med.	Low	Full
BOVA/SMAL	0.82	0.84	0.81	0.87	IVV/IWM	0.90	0.81	0.77	0.87
BOVA/IMAB	0.41	0.12	0.25	0.44	IVV/IEF	0.10	−0.07	0.16	−0.10
BOVA/FIXA	0.47	0.34	0.26	0.44	IVV/TIP	0.22	0.11	0.24	0.09
BOVA/DIVO	0.88	0.84	0.86	0.89	IVV/DVY	0.82	0.67	0.70	0.83
BOVA/FIND	0.84	0.82	0.84	0.79	IVV/FTXO	0.77	0.56	0.55	0.72
BOVA/MATB	0.59	0.56	0.59	0.69	IVV/VAW	0.87	0.77	0.75	0.86
BOVA/GOVE	0.94	0.89	0.92	0.91	IVV/ESGU	1.00	1.00	1.00	1.00
SMAL/IMAB	0.44	0.23	0.28	0.45	IWM/IEF	0.12	−0.08	0.14	−0.07
SMAL/FIXA	0.54	0.40	0.27	0.46	IWM/TIP	0.25	0.08	0.19	0.13
SMAL/DIVO	0.77	0.78	0.77	0.84	IWM/DVY	0.77	0.80	0.78	0.84
SMAL/FIND	0.79	0.70	0.71	0.71	IWM/FTXO	0.78	0.76	0.75	0.82
SMAL/MATB	0.35	0.45	0.38	0.56	IWM/VAW	0.84	0.82	0.81	0.87
SMAL/GOVE	0.81	0.80	0.77	0.79	IWM/ESGU	0.91	0.81	0.78	0.87
IMAB/FIXA	0.60	0.40	0.32	0.50	IEF/TIP	0.75	0.83	0.85	0.76
IMAB/DIVO	0.43	0.14	0.29	0.44	IEF/DVY	0.12	−0.15	0.06	−0.15
IMAB/FIND	0.34	0.14	0.27	0.36	IEF/FTXO	−0.04	−0.25	−0.13	−0.24
IMAB/MATB	0.18	0.03	0.10	0.31	IEF/VAW	0.12	−0.09	0.09	−0.11
IMAB/GOVE	0.41	0.18	0.23	0.42	IEF/ESGU	0.12	−0.06	0.17	−0.09
FIXA/DIVO	0.41	0.35	0.28	0.44	TIP/DVY	0.27	0.01	0.14	0.05
FIXA/FIND	0.45	0.32	0.22	0.36	TIP/FTXO	0.07	−0.09	−0.04	−0.03
FIXA/MATB	0.13	0.14	0.14	0.28	TIP/VAW	0.24	0.09	0.17	0.09
FIXA/GOVE	0.44	0.31	0.24	0.44	TIP/ESGU	0.23	0.12	0.26	0.09
DIVO/FIND	0.78	0.76	0.79	0.75	DVY/FTXO	0.82	0.87	0.81	0.87
DIVO/MATB	0.56	0.50	0.47	0.62	DVY/VAW	0.87	0.82	0.82	0.89
DIVO/GOVE	0.87	0.79	0.81	0.83	DVY/ESGU	0.81	0.65	0.68	0.82
FIND/MATB	0.34	0.27	0.36	0.40	FTXO/VAW	0.78	0.68	0.69	0.78
FIND/GOVE	0.84	0.73	0.78	0.72	FTXO/ESGU	0.77	0.55	0.54	0.71
MATB/GOVE	0.58	0.56	0.59	0.66	VAW/ESGU	0.86	0.76	0.75	0.85

Note: Correlations are computed conditional on volatility regimes classified by the HMM as High, Medium, and Low volatility, as well as over the full sample (2020–2024). Equity ETFs include broad market, factor, and sector exposures (Brazil: BOVA, SMAL, DIVO, FIND, MATB, GOVE; U.S.: IVV, IWM, DVY, FTXO, VAW, ESGU). Fixed-income ETFs include government bond proxies (Brazil: IMAB, FIXA; U.S.: IEF, TIP).

From a volatility regime perspective, there is a systematic increase in return dispersion during high-volatility states in both markets. In the U.S., these regimes are more clearly associated with lower mean returns, whereas in Brazil, the effect manifests predominantly through increased variability, without a monotonic pattern in average returns. Across all regimes, fixed-income ETFs maintain a relatively defensive posture, while equities, factors, and cyclical sectors exhibit higher sensitivity to regime transitions, reinforcing the rationale for regime-conditioned allocation strategies.

Table 7 reports the correlations between assets under high, medium, and low volatility regimes, as well as for the full sample. The results confirm evidence documented in the literature (Ang & Timmermann, 2012; Davoodi et al., 2024; Jacobs & Karagozoglu, 2014; Peng et al., 2022), indicating that the asset correlation matrix varies across regimes. This variation has direct implications for systemic risk dynamics and the effectiveness of portfolio diversification.

Asset correlations tend to increase in high-volatility environments, as noted by Kritzman et al. (2012) and Jacobs and Karagozoglu (2014). Such co-movements are highly relevant, as they imply that traditional diversification becomes less effective during periods of market stress, thereby limiting the potential for risk reduction.

Across regimes, the correlation structure reveals distinct dynamics between emerging and developed markets. In the Brazilian market, correlations rise during high-volatility periods, reflecting a pattern of contagion and a loss of diversification effectiveness, behavior typical of emerging economies that are more sensitive to liquidity shocks and sovereign risk. In contrast, in the U.S. market, specifically during the high-volatility regime, the correlation between equities (IVV) and treasuries (IEF) shifted to positive values (0.10, Table 7), breaking the traditional 'flight-to-quality' dynamic. This reflects the inflationary nature of the recent crisis (e.g., 2022), where both asset classes suffered. Consequently, rather than acting as a safe haven, Treasuries offered limited protection, explaining the dynamic strategy's reduced exposure to fixed income seen later in Table 8.

These results reinforce that regime-conditioned diversification, unlike traditional approaches, can generate gains over a full market cycle, corroborating the findings of empirical studies such as Oprisor and Kwon (2021) and Li et al. (2022). This occurs because a regime-based rebalancing strategy allows for dynamic adjustments that mitigate excessive losses during turbulent periods and optimize returns during market recoveries (Bae et al., 2014).

The regime-conditioned allocation strategy adopted in this study is based on the premise that both the distribution of asset returns and their dependency structure vary systematically across the volatility cycle. Under these circumstances, static allocation strategies calibrated on unconditional moments become particularly vulnerable to risk misspecification, especially during stress periods when correlations spike and tail risks dominate portfolio behavior.

In this context, the use of a Hidden Markov Model (HMM) allows for the probabilistic inference of the prevailing volatility regime, enabling dynamic adjustments to asset weights consistent with the dominant risk environment. A central aspect of the

Table 8
Average asset weights by volatility regime with a 5 bps transaction cost.

State	Brazilian ETFs								Total	
	BOVA	SMAL	DIVO	FIND	MATB	GOVE	IMAB	FIXA	Equity	Fixed
HighVol	0.000	0.017	0.409	0.036	0.243	0.000	0.278	0.016	0.705	0.295
MedVol	0.012	0.074	0.136	0.451	0.257	0.000	0.006	0.064	0.929	0.070
LowVol	0.017	0.171	0.093	0.392	0.326	0.000	0.000	0.001	0.999	0.000
	U.S. ETFs									
State	IVV	IWM	DVY	FTXO	VAW	ESGU	IEF	TIP	Equity	Fixed
HighVol	0.026	0.000	0.175	0.678	0.000	0.092	0.026	0.003	0.970	0.029
MedVol	0.131	0.079	0.212	0.343	0.003	0.031	0.188	0.013	0.798	0.201
LowVol	0.151	0.051	0.201	0.440	0.000	0.022	0.090	0.045	0.865	0.135

Note: Average weights calculated for the 2020–2024 period, under a long-only strategy with transaction costs $\gamma = 0.0005$ (5 bps). Equity ETFs include factor, sector, and broad market indices (BR: BOVA11, SMAL11, DIVO11, FIND11, MATB11, GOVE11; US: IVV, IWM, DVY, FTXO, VAW, ESGU), while fixed-income ETFs (Fixed) include government bonds (BR: IMAB11, FIXA11; US: IEF, TIP). Regimes are classified as low (LowVol), medium (MedVol), and high volatility (HighVol).

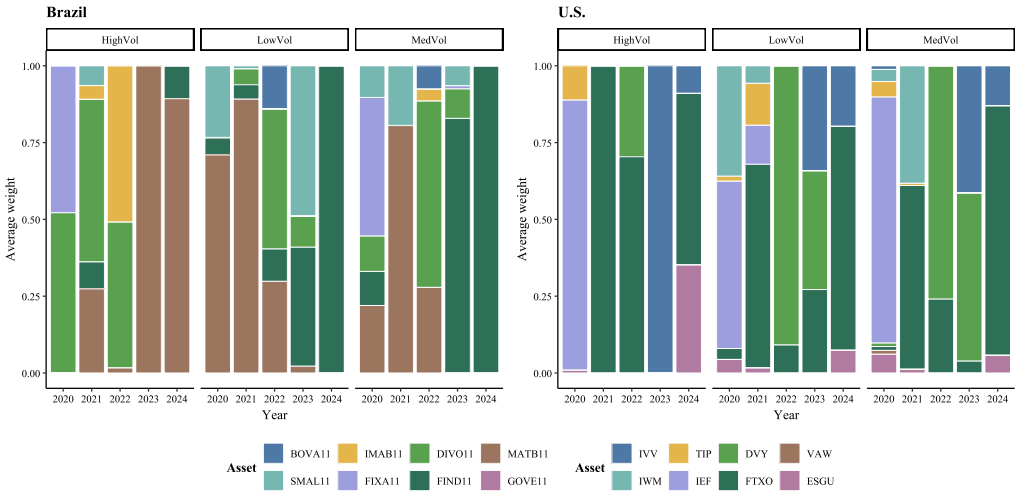


Fig. 4. Evolution of portfolio weights across volatility regimes by year.

proposed approach is that portfolio rebalancing does not occur at fixed intervals but only upon a change in the predicted regime. This imposes an operational discipline that balances adaptability with transaction cost control (Eq. (6)).

Table 8 reports the average asset weights under low, medium, and high volatility regimes. The results reveal economically consistent and heterogeneous patterns across the analyzed markets.

In the Brazilian market, the optimization results reveal a preference for real assets and equity exposure, even during stress periods. Contrary to a total flight-to-safety, the Total Equity allocation remained robust, ranging from 99.9% in the Low-Volatility regime to 70.5% in the High-Volatility regime (Table 8). However, a distinct defensive rotation occurs during high volatility: the model reallocates approximately 28% of the portfolio specifically into inflation linked government bonds (IMAB11), while reducing exposure to small caps (SMAL11) and broad market indices (BOVA11). This suggests a hedge against real interest rate risk rather than nominal risk.

This behavior suggests that during the analyzed period (2020–2024), the algorithm identified inflation-protected securities and specific equity sectors, such as Dividends (DIVO11, 40.9% in HighVol) and Basic Materials (MATB11, 24.3% in HighVol), as more effective hedges than nominal fixed income (FIXA11). This aligns with the domestic scenario of high inflation and the commodity boom that occurred during parts of the sample. The average weights across ETFs by regime and year are illustrated in Fig. 4.

In the U.S. market, the response to high volatility was characterized not by a flight to bonds, but by an aggressive sectoral rotation within the equity universe. As shown in Table 8, the Total Equity allocation actually increased to 97.0% in the High-Volatility regime, compared to 86.5% in Low-Volatility.

However, the portfolio composition shifted markedly: the model exited broad market exposure (IVV) and small caps (IWM), reallocating 67.8% to the Financial sector (FTXO) and 17.5% to Dividends (DVY). This seemingly counterintuitive adjustment reflects the specific dynamics of the post-pandemic recovery and monetary tightening cycle, in which financial stocks benefited from widening interest spreads, while Treasury bonds (IEF, TIP) faced correlation breakdowns and inflationary pressures, failing to serve as traditional safe havens.

Table 9
Performance comparison of strategies (Cost = 5 bps).

Market	Strategy	Cum. Ret.	Annual Ret.	ShR	MDD	Vol.
BR	Dynamic Regime	30.20	6.35	0.016	34.42	20.91
	Single Regime	15.70	3.46	−0.125	31.43	19.13
	Naive (EW)	**42.42**	**8.59**	**0.169**	**20.58**	**14.53**
US	Dynamic Regime	1.56	0.34	−0.003	36.18	21.46
	Single Regime	−18.49	−4.37	−0.266	43.34	17.85
	Naive (EW)	**50.95**	**9.41**	**0.631**	**21.70**	**14.23**

Note: Values extracted from the optimization with 8 ETFs and 3 regimes for the 2020–2024 period. Performance metrics, including the Sharpe Ratio (ShR), were computed using annualized excess returns and volatility based on daily log-returns (T = 252), following standard financial econometrics literature. The Israelsen (2005) refinement was applied to account for periods of negative excess returns caused by the high interest rate environment. MDD represents Maximum Drawdown and Vol volatility.

This behavior can be understood through the logic of the optimization model itself, which seeks to maximize expected returns while simultaneously minimizing portfolio risk and penalizing transaction costs (Eq. (6)). During low-volatility periods, although the perceived risk in equity markets is reduced, fixed-income ETFs may present a more favorable risk–return profile, particularly in economic contexts characterized by high real interest rates.

During the asset allocation period (2020–2024), the basic interest rate in the Brazilian market increased by 10.25 percentage points, while the U.S. market saw an increase of 4.5 percentage points. The upward movement of the base rate exerts downward pressure on corporate valuations, as it raises the cost of capital and reduces the present value of future cash flows, leading to lower stock prices and a subsequent reduction in market capitalization (Kim, 2003). On the other hand, interest rate hikes increase the profitability of fixed-income securities without necessarily altering their perceived risk, making them relatively more attractive in terms of the risk-return trade-off. By capturing these favorable conditions, the model naturally increases allocation to these assets.

Furthermore, in high-volatility periods, while a higher allocation to fixed-income ETFs might be expected, suggesting a demand for predictability and safety, the model may identify specific return opportunities in equity assets, especially after significant price corrections. Paradoxically, the portfolio may, on average, show lower weights in fixed-income ETFs during these periods, suggesting that the model seizes specific moments to capture short-term equity gains. Additionally, the model consistently balances the return–risk trade-off against the welfare loss associated with excessive portfolio turnover.

It is worth noting that the magnitude of the interest rate hike in Brazil was significantly larger than in the U.S., justifying the high average concentration in fixed-income ETFs recommended by the model, regardless of the volatility regime. In the U.S., the milder interest rate cycle allowed the model more flexibility to alternate between ETFs.

Although the sharp rise in the Brazilian interest rate (Selic) significantly raised the opportunity cost of equity, the optimizer avoided nominal fixed income (FIXA11), likely due to the duration risk associated with the opening of the yield curve. Instead, the allocation to IMAB11 in the high-volatility regime captures the demand for protection against real interest rate fluctuations. The persistence of high equity weights can be attributed to the specific composition of the chosen ETFs (e.g., Materials and Dividends), which served as effective proxies for real assets during the inflationary spike.

Consistent with Chauvet et al. (2015), these results suggest a relationship between financial volatility and macroeconomic fluctuations, reinforcing the importance of a detailed analysis of the macroeconomic and financial conditions associated with each regime. They also highlight the optimization model's ability to identify relevant nuances for dynamic allocation, surpassing traditional heuristic patterns like the 60/40 split frequently mentioned in the literature (Grant et al., 2024; Kandhari, 2024).

These results indicate that regime-conditioned allocation is not merely a binary "risk-on/risk-off" strategy, but rather reflects the relative risk and return conditions specific to each market and volatility regime. Contrary to initial expectations that active regime-based management would offer superior tail protection, Table 9 reveals that the Naive (1/N) strategy yielded the lowest Maximum Drawdown (MDD) in both markets (20.58% in Brazil and 21.70% in the U.S.). The underperformance of the dynamic strategy relative to the naive benchmark is consistent with DeMiguel et al. (2009) and Davoodi et al. (2024), who argue that estimation errors in dynamic optimization can erode gains in highly uncertain environments.

While the dynamic strategy effectively reduces downside risk in the U.S. market, lowering the Maximum Drawdown from 43.34% under the Single Regime strategy to 36.18%, its impact in the Brazilian market is more nuanced. In Brazil, dynamic reallocation leads to a slightly higher drawdown (34.42% versus 31.43%), reflecting greater exposure to regime transitions in a high-volatility environment. However, this additional downside risk is offset by a faster and more consistent recovery, resulting in a positive Sharpe Ratio, in contrast to the negative risk-adjusted performance of the static strategy.

Consistent with this evidence, the Dynamic Regime strategy outperforms the Single Regime approach in both cumulative returns and risk-adjusted terms in the Brazilian market (Table 10). Even under elevated transaction costs of 20 bps, cumulative returns remain higher for the dynamic strategy (19.09% versus 14.97%). Although the relatively high turnover (59.59 reallocations over the sample period) increases sensitivity to transaction costs, this effect does not outweigh the informational advantage derived from regime identification.

In the U.S. market, the margin for economic gain is narrower. Nonetheless, although the dynamic strategy generated a negative Sharpe ratio (−0.003), it significantly outperformed the static strategy (which suffered structural losses near 18%), highlighting its role in capital preservation rather than yield generation. In absolute terms, the dynamic strategy's performance turns negative

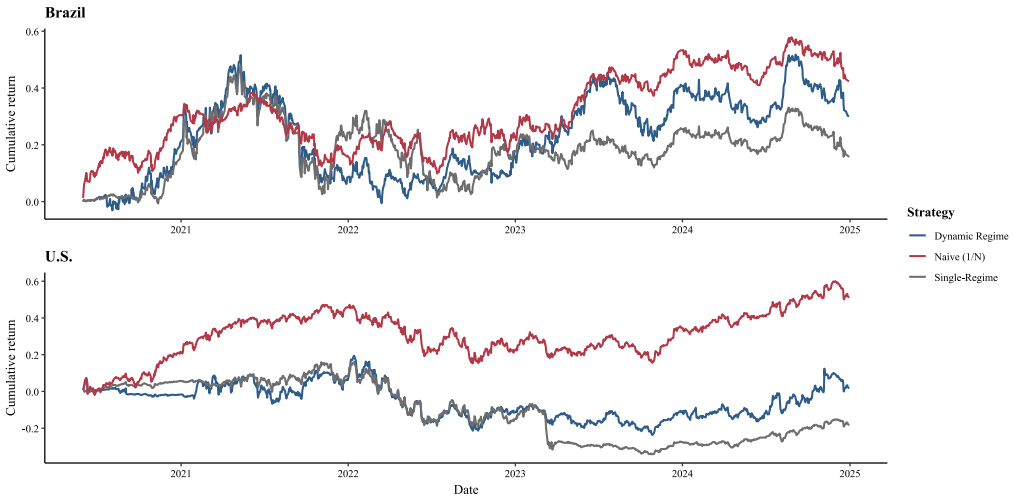


Fig. 5. Cumulative performance of portfolio strategies over the out-of-sample period. The figure reports the wealth index for the Dynamic Regime strategy, the Single-Regime static strategy, and the naive equal-weighted benchmark (1/N) in Brazil and the U.S. The comparison highlights the trade-off between dynamic risk adaptation, capital preservation during downturns, and long-run cumulative performance.

Table 10
Performance sensitivity to transaction costs (2020–2024).

Market	Cost (bps)	Dynamic Regime		Single Regime (Static)	
		Cum. Ret.	Sharpe	Cum. Ret.	Sharpe
BR	0	34.12%	0.049	15.94%	−0.123
	2.5	32.15%	0.032	15.82%	−0.124
	5.0	30.20%	0.016	15.70%	−0.125
	10.0	26.39%	−0.017	15.46%	−0.128
	20.0	19.09%	−0.083	14.97%	−0.133
US	0	3.68%	0.018	−18.26%	−0.263
	2.5	2.61%	0.008	−18.38%	−0.264
	5.0	1.56%	−0.003	−18.49%	−0.266
	10.0	−0.53%	−0.024	−18.73%	−0.269
	20.0	−4.58%	−0.066	−19.20%	−0.276

Note: Values extracted from optimization with $\lambda = 1$. Total Turnover represents the sum of absolute weight changes over the entire period.

when transaction costs approach 10 bps, indicating that its effectiveness depends on efficient execution and low trading friction, conditions that seem realistic for highly liquid ETFs like IVV and IEF.

Fig. 5 complements the portfolio performance tables by depicting the full out-of-sample trajectory of the Dynamic Regime, Single-Regime, and Naive strategies. The figure illustrates that the regime-based approach is best interpreted as a risk-management device rather than a universal return-enhancing rule, since its primary advantage lies in mitigating severe losses relative to the static benchmark, while the naive portfolio remains highly competitive in cumulative return terms (De Jong, 2018).

The choice of 5 bps (0.05%) as the base cost in the primary analysis proves to be conservative and methodologically appropriate. This value penalizes the dynamic model's turnover without compromising its ability to mitigate losses associated with abrupt regime shifts. For institutional investors operating with lower costs, the strategy's potential gains would be even more significant, reaching, for example, cumulative returns of 34.12% in Brazil and 3.68% in the United States in a zero-cost scenario.

The challenge of generating positive excess returns in the Brazilian market is consistent with the findings of Campani et al. (2021). The authors characterize the local risk-free rate not merely as a theoretical safe haven, but as a high-yield investment vehicle that competes aggressively with equity allocations.

Consequently, the structural elevation of this "hurdle rate" compresses equity risk premiums, explaining the negative Sharpe ratios observed across most scenarios during the monetary tightening cycle of 2020–2024. In this context, the value of the dynamic HMM strategy lies in its ability to mitigate drawdowns (risk management) rather than solely in maximizing absolute returns against an exceptionally high benchmark.

Taken together, these results indicate that the primary contribution of regime-conditioned allocation lies not in maximizing unconditional returns, but in mitigating the structural failures of static optimization in environments subject to persistent shifts in volatility and correlation. In markets characterized by prolonged trends and low parametric uncertainty, simple strategies like the $1/N$ portfolio may remain competitive. However, in contexts marked by volatility clustering, macroeconomic shocks, and structural

instability, as observed during the 2020–2024 period, the HMM-based approach offers relevant economic gains by aligning risk exposure with prevailing market conditions.

Thus, the proposed strategy is particularly suitable for investors seeking a balance between dynamic market adaptation and intertemporal risk control, avoiding both the rigidity of static optimization and the unrestricted exposure of purely naive strategies.

5. Conclusion

This study evaluated the effectiveness of alternative volatility proxies (historical, conditional, and implied) in identifying market regimes via Hidden Markov Models (HMM) and their subsequent economic value in asset allocation strategies. The empirical evidence supports conditional volatility (GARCH as the superior proxy for regime identification in both the Brazilian and U.S. markets). Unlike implied volatility, which exhibited excessive sensitivity to short-term noise and threshold variations, the GARCH-based specification provided the greatest parameter stability and classification robustness, essential attributes for operationalizing dynamic portfolios.

A key finding of this research is the structural identification of three volatility regimes (low, medium, and high). Contrary to binary specifications often assumed in the literature for interpretability, the Bayesian Information Criterion (BIC) results demonstrated that a three-state model better captures the complex dynamics of financial markets, specifically identifying transitional phases that binary models fail to detect. This granular identification allowed for a more precise assessment of international risk transmission. Our Granger causality tests, applied to logit-transformed regime probabilities, confirmed that the U.S. high-volatility regime acts as a statistical precursor to regime shifts in Brazil. This lead–lag relationship validates the "Early Warning System" hypothesis, suggesting that emerging markets import latent uncertainty perceptions from core markets before they fully materialize in local asset prices.

From an investment perspective, the results highlight the distinct roles of regime-based strategies. The Dynamic Regime strategy consistently outperformed the traditional Static Mean-Variance (Single Regime) strategy in risk-adjusted metrics, successfully mitigating severe drawdowns, most notably in the U.S. market, where the static strategy suffered structural losses (−18.49%) while the dynamic strategy preserved capital (+1.56%). However, the dynamic strategy did not outperform the Naive (1/N) benchmark in terms of total cumulative return. This finding is consistent with the "estimation error" puzzle documented by DeMiguel et al. (2009): in highly correlated environments with elevated transaction costs, the benefits of dynamic timing are often offset by the robustness of simple diversification. Thus, HMM-based allocation is best interpreted as a sophisticated risk management tool for downside protection rather than a universal yield enhancement mechanism.

Finally, while the bilateral Brazil–U.S. design ensured data quality and liquidity for the transaction cost analysis, it represents a limitation in geographic scope. Future research should expand this framework to a broader panel of emerging economies to test the universality of the U.S. volatility spillover. Additionally, incorporating macroeconomic variables (e.g., yield curve slopes, credit spreads) as exogenous drivers of transition probabilities could bridge the gap between statistical regime identification and fundamental economic drivers, offering a promising avenue for improving the predictive accuracy of dynamic allocation models.

Data availability

The data and code supporting the findings of this study are openly available at https://doi.org/10.5281/zenodo.15649071.

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